
RL2Ocean Calibration and Validation ATBD

Release 1.0

RL2Ocean consortium

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Note

The source abstract contains only Lorem ipsum placeholder text and is not reproduced. The source's substantive content and section outline are preserved; headings without body text are identified in the relevant chapters rather than filled with invented material.

CHAPTER ONE

INTRODUCTION

1.1 Purpose and scope

The Copernicus Programme, established by the European Union, is a framework for Earth Observation and Monitoring, designed to support environmental protection, civil protection, and civil security efforts. At the heart of this programme is the Copernicus Space Component (CSC), which comprises the Sentinel missions. These missions are integral to providing essential data across various thematic domains, thereby facilitating informed decision-making and policy development.

One of the new advancements within this framework is the Copernicus Radar Observing System for Europe – L-Band (ROSE-L). This initiative is aimed at enhancing existing capabilities by introducing L-band Synthetic Aperture Radar (SAR) data. The L-band SAR is set to complement the C-band radar data provided by Sentinel-1, thus addressing specific observational needs within environmental and maritime surveillance.

1.2 Cal/Val context

Quality Assessment (QA) is required for remote sensing and Earth Observation (EO) instruments in order to produce reliable measurements and ensure accuracy. Quality of products are commonly assessed through a Calibration and Validation process (Cal/Val). Consequently, the development of the prototype and operational ROSE-L Ocean products processors (L2O-GPP and L2O-IPF respectively) must be followed by the validation of products. A validation strategy is meant to be planned prior to calibration and validation activities. The required strategy shall cover different stages of the processor and different phases of ROSE-L mission, from pre-launch (phase D) to exploitation (phase E2). The strategy shall include methodologies, datasets, performance and uncertainty metrics.

The ROSE-L instrument will operate in scanning synthetic aperture radar (ScanSAR) mode. The system will support Dual-Pol, Quad-Pol and Wave modes. The Cal/Val strategy is also meant to cover all modes and polarizations. ROSE-L will not be the first SAR system functioning within L-band. ALOS, ALOS-2, ALOS-4, SAOCOM and NiSAR are, or have been, operating in L-band too. However, there are differences between those systems and the ROSE-L system, e.g. incidence angle, range bandwidth, swath width, among others. Furthermore, ROSE-L incorporates the use of a new technique, the Scan-On-Receive (SCORE) technique with Multiple Azimuth Phase Centers (MAPS). The different instrument design will impose differences in the retrievals at low level which are meant to influence the inversion into L2 products.

Regarding the development of a retrieval algorithm but also Cal/Val activities, instrument differences between ROSE-L and previous SAR L-band instruments impose a challenge; the need of producing and validating ocean products from ROSE-L scenes during the pre-launch phase. The lack of real ROSE-L data will be mitigated by the use of similar acquisitions by the above-mentioned systems; although this introduces a non-negligible degree of uncertainty. For Cal/Val purposes, simulated ROSE-L scenes will be produced. The achievable accuracy of simulated ROSE-L will also be analysed during the In-Orbit-Commissioning (IOC) phase, E1. Once real ROSE-L data is available, the outputs from the L2O-IPF processor will need to be validated again, with the possibility to encounter a need for a readjustment of the processor or introduction of supplementary calibration steps.

A tool for assessing the quality of the ocean products is vital. The Cal/Val tool, L2O-QC, is meant to be developed in parallel to the development of the ocean processors, L2O-GPP and L2O-IPF. The L2O-QC software is required to be able to ingest datasets of different nature; ROSE-L acquisitions, reference datasets (either in-situ or remote), and auxiliary data. The goal is to collocate the different datasets in order to undertake the quality assessment, where the accuracy and reliability of the ROSE-L L2 ocean products will be determined and ensured. The underlying

objective is to provide users with a set of performance and uncertainty metrics necessary for a deep understanding of the products and subsequent appropriate use and application.

1.3 L2 product scope

The L2 products to validate are already listed in the objectives of L2O-GPP and L2O-IPF: Ocean Surface Vector Stress (OSVS), Sea State, and line-of-sight (LoS) surface currents. Within each of the three thematic products, the following L2 variables are meant to be produced and therefore validated:

- OSVS:
 - Wind vector
 - Wind stress
 - Hub height wind speed
- Sea state:
 - Total significant wave height
 - Mean period wind-sea
 - First moment wave period
 - Second moment wave period
 - Wave height swell dominant system
 - Wave height swell secondary system
 - Significant wave height wind-sea
 - Wave direction
 - Wave directional energy spectrum
- LoS currents:
 - LoS wave Doppler
 - LoS surface current component

1.4 Cal/Val objectives and performance

The key objectives of the Calibration and Validation (Cal/Val) phase include ensuring that all L2 geophysical products meet internationally recognized standards, such as SI traceability, and align with QA4EO principles to guarantee high data quality. Cal/Val aims to validate the accuracy, precision, and stability of L2 products using independent datasets, such as Fiducial Reference Measurements (FRM), and cross-sensor harmonization. A critical focus is on quantifying and documenting uncertainty propagation from instrument measurements to retrievals while ensuring the retrieval algorithms are robust under diverse environmental conditions.

To meet performance requirements, L2 products must achieve predefined thresholds in accuracy, precision, and stability, with uncertainties traceable and reliable across pixel and aggregate levels. Products must demonstrate consistency over full mission coverage, addressing global, regional, and temporal scales, and maintain defined operational latencies (e.g., near-real-time or reprocessed outputs). Validation activities use measurable metrics, including statistical bias, RMS error, and spatial-temporal coverage rates, to assess product quality systematically. Lastly, the outputs should be delivered in user-ready formats with comprehensive documentation while maintaining version control and data provenance.

1.5 ATBD purpose

The aim of the Calibration and Validation Algorithm Theoretical Basis Document (ATBD) is to provide a comprehensive and transparent description of the scientific principles, mathematical methods, and processing steps used to generate ROSE-L L2 ocean products. It ensures that the algorithms are well-documented, reproducible, and

traceable, facilitating a clear understanding of how measurements are transformed into geophysical quantities. The ATBD serves as a foundation for ensuring data quality, guiding calibration/validation efforts, and supporting user confidence in the satellite mission's outputs.

1.6 Applicable and reference documents

1.7 Acronyms

CANVAS

CALibration aNd VALidation for Sar.

CEOS

Committee on Earth Observation Satellites.

DC

Doppler Centroid.

DCA

Doppler Centroid Anomaly.

DORIS

Doppler Orbitography and Radiopositioning Integrated by Satellite.

EO

Earth observation.

ESA

European Space Agency.

EU

European Union.

FDR

Fundamental Data Record.

FIDUCEO

Fidelity and Uncertainty in Climate Data Records from Earth Observation.

FP7

Seventh Framework Programme.

FRM

Fiducial Reference Measurement.

GEO

Group on Earth Observations.

GEOSS

Global Earth Observation System of Systems.

GUM

Guide to the expression of Uncertainty in Measurement.

JCGM

Joint Committee for Guides in Metrology.

QA4EO

Quality Assurance Framework for Earth Observation.

SAR

Synthetic Aperture Radar.

SI

International System of Units.

TDP

Thematic Data Product.

UTD

Uncertainty Tree Diagram.

VIM

International Vocabulary of Metrology.

1.8 Definitions and conventions

Accuracy

The quality of being near to the true value. The source marks this definition as requiring a reference.

Comparison

Metrologists validate uncertainty analysis and confirm traceability through comparisons. The Mutual Recognition Arrangement requires regular, formal international comparisons to establish the degree of equivalence between SI realisations and disseminations in different countries.

Dataset

A combination of data records, such as numerical values needed to analyse and interpret a natural process, and the related information content.

Fiducial Reference Measurement

A suite of independent, fully characterised, and traceable sub-orbital measurements, relevant to space-based observations, following the guidelines of the GEO/CEOS Quality Assurance Framework for Earth Observation (QA4EO).

Fundamental Data Record

A record, of sufficient duration for its application, of uncertainty-quantified sensor observations calibrated to physical units and located in time and space, together with ancillary and lower-level instrument data used to calibrate and locate the observations and to estimate uncertainty (*Woolliams et al. 2025a*).

Interoperability

The ability of data or tools from non-cooperating resources to integrate or work together with minimal effort (*Wilkinson et al. (2016)²*).

Interoperable

Describes data or tools from non-cooperating resources that can integrate or work together with minimal effort (*Wilkinson et al. (2016)³*).

Measurand

The quantity intended to be measured, and the single output quantity from a measurement model.

Metrology

The science of measurement and its application (*BIPM et al., n.d.-b*).

Thematic Data Product

A record, of sufficient duration for its application, of uncertainty-quantified retrieved values of a geophysical variable, along with ancillary data used in retrieval and uncertainty estimation.

Traceability

A property of a measurement relating its value to a stated metrological reference through an unbroken chain of calibrations or comparisons, with uncertainties evaluated and methodologies documented at each step.

Uncertainty analysis

The review of uncertainty sources and their propagation through the traceability chain, based on the suite of documents prepared by the JCGM and known as the GUM.

² <https://doi.org/10.1038/sdata.2016.18>

³ <https://doi.org/10.1038/sdata.2016.18>

**CHAPTER
TWO**

PROCESSING MODEL AND DATA FLOW**2.1 Pre-launch phase (D)**

During the pre-launch phase (phase D), activities focus on preparing and validating the L2 processing chain, ensuring readiness for on-orbit operations. Key tasks include finalizing L2 requirements, algorithms, and performance budgets. Simulators and proxy datasets are leveraged to validate retrieval algorithms and uncertainty models. Pre-launch instrument characterization is integrated into the processing chain to ensure retrievals are robust to on-orbit conditions. A comprehensive Cal/Val Plan is developed, identifying validation sites, protocols, data matchups, and metrics.

2.2 Commissioning phase (IOC or E1)

In the commissioning phase (IOC or phase E1), the primary goal is to validate L2 products using on-orbit data and release initial beta/provisional products with documented performance. Early activities include confirming L1 performance, performing vicarious calibrations, and matching L2 products with ground-based and airborne reference datasets (FRM sites, buoys, ship-based campaigns, etc.). Algorithms are fine-tuned using real-time data, and initial uncertainty models are validated. Early reprocessing cycles are conducted to improve product stability, and performance against key metrics is assessed. Products are released with quality summaries, and ongoing monitoring dashboards are set up.

2.3 Exploitation phase (E2)

The exploitation phase (phase E2) ensures L2 products maintain accuracy, precision, and stability over the mission lifecycle. Routine validation is conducted using reference networks, continuous KPI monitoring, and periodic reprocessing cycles for algorithm and calibration improvements. Traceability to standards (QA4EO, FRM) and uncertainty validation is maintained, while long-term drift corrections support stable time series. Community engagement, user feedback, and updates to documentation (validation reports, product guides) are prioritised. The goal is to achieve validated product maturity, ensuring high-quality data for science and operational applications.

2.4 End-to-end data flow

Processing-flow diagram, processing-step mapping, or algorithm identifiers still missing.

CHAPTER**THREE**

L-BAND SAR DATASETS DESCRIPTION**3.1 Generated datasets****3.1.1 Reconditioned****3.1.2 Emulated****3.1.3 Simulated****3.2 Third-party datasets****3.2.1 ALOS****3.2.2 ALOS-2****3.2.3 ALOS-4****3.2.4 SAOCOM****3.2.5 NiSAR**

CHAPTER

FOUR

AUXILIARY DATASETS DESCRIPTION

CHAPTER FIVE

REFERENCE DATASETS DESCRIPTION

In L2o-qc, reference data are used for both calibration and validation of SAR-derived geophysical parameters. The L2 calibration process is exclusively implemented through Numerical Weather Prediction Ocean Calibration (NOC), which is specifically applied to the wind vector under the OSVS thematic area. Validation is performed against multiple independent reference sources for all thematic areas. The following sections describe the reference datasets used for calibration and validation purposes.

5.1 Ocean Surface Vector Stress

5.1.1 Wind vector

5.1.1.1 C-band SAR

Sentinel-1 L2 OCN - OWI

Description:

Ocean Wind field (OWI) component of the Sentinel-1 Level-2 Ocean (L2_OCN) product, providing 10 m neutral wind speed and direction on a regular grid, derived from the Level-1 SAR backscatter using a CMOD5.N geophysical model function. The OWI component is available in Sentinel-1 Interferometric Wide (IW), Extra Wide (EW), and Stripmap (SM) acquisition modes.

In general, L1 and L2 Sentinel-1 products are generated by the Instrument Processing Facility (IPF) and carry an `ipf_version` with them to distinguish between different processing versions.

Characteristics:

land pixels are masked in the source product. The OWI product is typically delivered at an 1x1 km grid for all acquisition modes. The nominal swath widths are mode dependent with IW $\sim$ 250 km, EW $\sim$ 400 km, and SM $\sim$ 80 km.

Data quality:

The OWI product carries a `wind_quality` flag and an `inversion_quality` flag. `inversion_quality` is related to the inversion, which depends on the NRCS, incidence angle, azimuth look angle and ancillary wind speed and direction. The `inversion_quality` flag indicates the consistency between SAR and ancillary winds. The `wind_quality` flag is more general including both the inversion quality as well as the geophysical quality for the whole product.

The `inversion_quality` flag has the following values:

- 0: high quality
- 1: medium quality
- 2: low quality

The `wind_quality` flag has had several different classifications. Currently (`ipf_version` = 4.03) it ranges from 0 – 4, where:

- 0: no data
- 1: bad

- 2: suspect
- 3: acceptable
- 4: good.

In the past (`ipf_version` < 4.03), its range ran from 0 – 3, where:

- 0: high quality
- 1: medium quality
- 2: low quality
- 3: bad quality.

To avoid interpreting the wrong meaning, `wq_flag_meanings` is read from the metadata before filtering.

Only high/medium (acceptable/good) flags are accepted. In addition, data points are rejected when both flags are NaN.

Availability:

3 April 2014 - present with global coverage (6 days revisit rate).

Dataset acquisition:

Copernicus Data Space Ecosystem (<https://dataspace.copernicus.eu/>); OData catalogue/download API at <https://catalogue.dataspace.copernicus.eu/odata/v1/Products>; free registration required.

RADARSAT-2 Wind

Description:

An independently-derived high-resolution sea surface wind speed product distributed by NOAA/NCEI, derived from RADARSAT-2 C-band SAR backscatter. Wind speed at 10m height is retrieved by inverting the CMOD4 geophysical model function.

Characteristics:

wind speed only (sub-kilometer pixel resolution). The accompanying direction field is the NWP-model direction used to resolve the CMOD4 180° ambiguity, not an independently retrieved SAR direction.

Two distinct filename/metadata eras exist in the archive (pre- and post-2024).

Data quality:

Post-2024 RADARSAT-2 wind files carry `pixel_level_quality_flags` parameter that can be used to mask land/ice/quality. A cell is deemed valid when this flag equals 5 (“valid wind in valid water region”). Pre-2024 files only contain a coarser mask/ice-mask combination, which is used as fallback `mask==1 & icemask==1`. All other cells are given NaN as value.

Availability:

May 2014 - present (~ 1 month latency). The available data via NOAA is heavily centered around the polar regions with most occurrences near Alaska and the Bering strait as can be seen in Figure Fig. 5.1.

Dataset acquisition:

NOAA NCEI THREDDS catalogue, <https://www.ncei.noaa.gov/thredds-ocean/catalog/sar-winds/radarsat2/catalog.html> (public, no authentication); per-granule bounding boxes are available via THREDDS’ NCML metadata service, avoiding full-file downloads for non-overlapping scenes.

5.1.1.2 Numerical Prediction Model

ERA5 Reanalysed wind

Description:

hourly, global 10 m wind field from ECMWF’s ERA5 atmospheric reanalysis, produced under the Copernicus Climate Change Service (C3S) and distributed via the Copernicus Climate Data Store (CDS). Used as an independent,



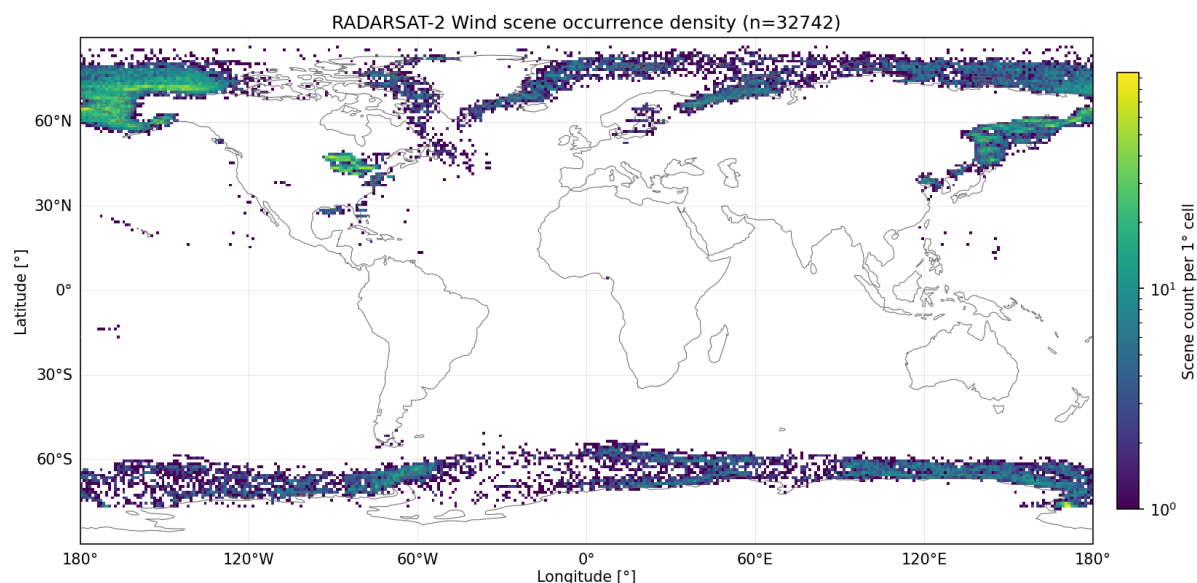


Fig. 5.1: Coverage of RADARSAT-2 high resolution sea surface winds via the NOAA/NCEI THREDDS archive. The center locations of all 32742 available RADARSAT-2 products from May 2014 - July 2026 have been plotted.

model-based reference wind field, collocated via bilinear spatial and hyperbolic temporal interpolation rather than point/swath matching.

Characteristics:

global regular latitude/longitude grid at 0.25° spacing, hourly output. The 10 m zonal and meridional wind components (u_{10} , v_{10}) are retrieved from the `reanalysis-era5-single-levels` dataset, together with a land-sea mask (lsm); wind speed and direction are derived after interpolation to the query location.

Data quality:

grid cells/query points with `lsm > 0.5` (land) are excluded before collocation. ERA5 is documented and evaluated in (Hersbach et al. 2020); an independent characterisation of its 10 m wind field (Belmonte Rivas and Stoffelen 2019) reports mean wind biases of up to 0.5 m/s in the zonal and meridional components, stable over time (interannual variability ~ 0.1 m/s), and roughly 20% lower instantaneous RMS wind speed error than the preceding ERA-Interim reanalysis.

Availability:

1940 - present with global coverage. A preliminary near-real-time extension (ERA5T, ~ 5 day latency) is later superseded by the quality-controlled final release (~ 3 month latency).

Dataset acquisition:

CDS dataset `reanalysis-era5-single-levels`, via the `cdsap` Python client; requires free CDS registration, with the API key stored in `~/.cdsapirc`.

5.1.1.3 In-Situ Observations

ECMWF MARS archive

Description:

Buoy observations relayed onto the WMO Global Telecommunication System (GTS) and archived by ECMWF's Meteorological Archival and Retrieval System (MARS), retrieved with `obstype=181/182` (BUFR moored and drifting buoys respectively). Relevant variables are near-surface wind speed (WSPD) and direction (WDIR). As default, `12o-qc` validates against GTS. Copernicus Marine in-situ data (see [Copernicus Marine](#)) can be used as alternative or a waterfall combining both: the waterfall downloads both sources for the full recipe bounding box and window, then at collocation-tree build time drops any Copernicus Marine station whose platform ID GTS already reported for that window, preferring GTS's value wherever GTS itself reports.

Characteristics:

point observations at native platform sampling rate ranging from once every 3-hrs to every 6-minutes. Default collocation in l2o-qc uses the point vs layer scheme with aggregation windows of ± 30 min / 5 km default radii (user adjustable via the .yaml recipe file).

Data quality:

Individual GTS BUFR reports carry no per-observation quality flag, because GTS requires the data processing center that owns a buoy to run automatic real-time quality control before an observation is transmitted onto the GTS (Panel 2011). The mandated real-time checks include, among others:

- Gross error checks against fixed physical limits. For wind, 0–100 m/s (speed) and 0–360° (direction).
- Climatological limit checks against operator- or International Comprehensive Ocean-Atmosphere Data Set (ICOADS)-supplied bounds for the platform's location and season.
- Land/sea test; any observations that appear on land are rejected.
- Location sanity checks (rejecting (0, 0), $\pm 180^\circ$, $\pm 90^\circ$ sentinel coordinates; a drifting buoy's position older than 48 hours, or a moored buoy's position older than its own (re-)deployment age).
- Checksum/Cyclic Redundancy Checks (CRC) validation of the transmission link between platform, satellite, and ground station.
- Duplicate/near-duplicate observation suppression, and cascading rejection when an associated depth level or compensating sensor itself fails QC.

In addition to the quality checks performed by the owners of the platforms before ingestion into the GTS database, there are delayed mode quality control guidelines in place. Principle meteorological and oceanographic centers, including ECMWF, compare buoy reports against NWP model analysis to check for erroneous or biased data, feeding corrections back to the platform operators (Panel 2011).

Availability:

The ECMWF MARS archive has archived data from 28-11-2016 - present.

Dataset acquisition:

ECMWF MARS archive (<https://www.ecmwf.int/en/forecasts/access-forecasts/access-archive-datasets>), obstype=181/182 (BUFR moored and drifting buoys), retrieved via the `ecmwf-api-client` Python package (`ecmwfapi.ECMWFService`); credentials read from `~/.ecmwfapirc` (ECMWF's standard Web API key file), obtained through an ECMWF Member/Co-operating-State-affiliated account. The request shape is day-based and global, meaning that each retrieval contains all wind, wave, and currents sections a given buoy reported for that day. Data is provided in daily BUFR files.

Copernicus Marine**Description:**

Dataset ID INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030, which contains in-situ observations. Relevant platforms reporting near-surface wind speed (WSPD) and direction (WDIR) are moored buoys, drifting buoys, ferryboxes, and tidal gauges.

Characteristics:

point observations at native platform sampling rate, which is typically hourly or every 10-minutes. Default collocation in l2o-qc uses point vs layer collocation. The aggregation windows, which are user adjustable via the .yaml file, utilize ± 30 min / 5 km aggregation radii.

Data quality:

Copernicus Marine in situ Thematic Assembly Centre (TAC) has no commitment to manage in situ meteorological observations. Reported quality flags are kept in the NetCDF files, with no additional quality control (Wehde et al. 2025). The QC flag ranges from 0 – 9 (Copernicus In Situ Tac Data Management Team 2025):

- 0: No QC was performed
- 1: Good data



- 2: Probably good data
- 3: Bad data although potentially correctable
- 4: Bad data
- 5: Value changed
- 6: Value below detection/quantification
- 7: Nominal value (e.g. reported rather than observed value like instrument target depth)
- 8: Interpolated value
- 9: Missing value

Due to the lack of additional quality control, buoy data from the ECMWF MARS archive is preferred.

Availability:

Platform dependent, the Copernicus marine environment monitoring services (CMEMS) dataset provides both historical data per platform and recent data. Not all platforms have both WSPD and WDIR available. Geographical coverage in 2026 can be seen in Figure Fig. 5.2.

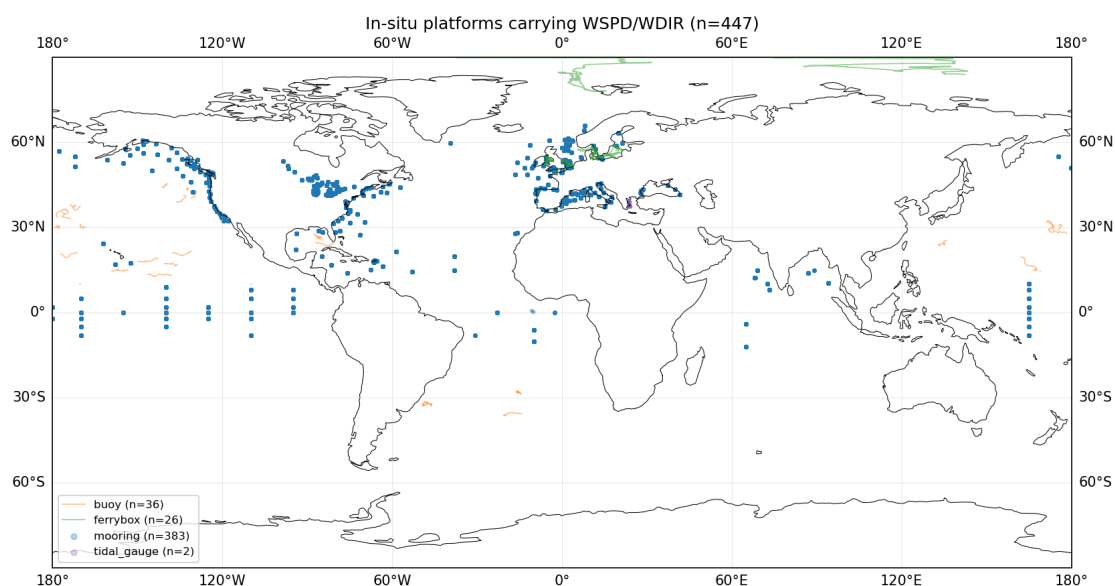


Fig. 5.2: Coverage of Copernicus Marine in-situ platforms that provided wind speed and direction during the first half of August 2026.

Dataset acquisition:

Copernicus Marine in-situ dataset (https://data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/), download via the copernicusmarine Python client (free registration).

5.1.1.4 Scatterometers

ASCAT

Description:

L2 product of 10 m neutral wind speed and direction wind-vector cells (WVCs) retrieved from C-band backscatter measured by the Advanced Scatterometer (ASCAT) aboard MetOp-B and MetOp-C, processed and distributed by EUMETSAT's Ocean and Sea Ice Satellite Application Facility (OSI-SAF) as product OSI-104 ("ASCAT Wind Product").

Characteristics:

dual fan-beam swath geometry, two ~ 550 km wide swaths either side of the sub-satellite track separated by a ~ 360 km nadir gap; wind-vector cells are provided on a regular along-/across-track grid at 12.5 km spacing. Data are distributed as 2-D swath NetCDF files and flattened to a point-geometry representation before collocation.

Data quality:

each WVC carries a bitmask `wvc_quality_flag`; cells for which any of the following bits are set are rejected before collocation:

- some portion of the WVC is over land
- some portion of the WVC is over ice
- wind inversion not successful
- not enough good σ_0 measurements for wind retrieval
- distance to the geophysical model function (GMF) too large

Furthermore, the OSI-SAF ASCAT (Metop-B/C) global 25 km and coastal 12.5 km wind products have been validated using buoys and triple collocation (Verspeek et al. 2019), showing wind quality well within the OSI-SAF product requirements (< 2 m/s wind component standard deviation, < 0.5 m/s speed bias). Against buoys, speed bias of 0.10/0.08 m/s and u-/v-component standard deviations of 1.62/1.90 (Metop-B) and 1.59/1.89 m/s (Metop-C) were found at 12.5 km. Triple collocation gives scatterometer-perspective errors of 0.62/0.78 (Metop-B) and 0.67/0.78 m/s (Metop-C) in the u-/v-components at 12.5 km (Verspeek et al. 2019).

Availability:

MetOp-B (launched 2012) and MetOp-C (launched 2018) provide continuous global coverage.

Dataset acquisition:

EUMETSAT Data Store, collection `E0:EUM:DAT:METOP:OSI-104`, via the `eumdac` Python client (free registration).

HY-2B/HY-2C

Description:

10 m wind speed and direction wind-vector cells retrieved from Ku-band backscatter measured by the scatterometers aboard Haiyang-2B (HY-2B) and Haiyang-2C (HY-2C) ocean satellites (CNSA/NSOAS), reprocessed into the same OSI-SAF wind-product format/flag conventions as ASCAT and redistributed via KNMI's OSI-SAF wind FTP server.

Characteristics:

HY-2B/HY-2C carry a Ku-band, wide-swath (~ 1700 km) pencil-beam scatterometer; wind-vector cells are provided at 25 km spacing/resolution.

Data quality:

the same `wvc_quality_flag` bitmask and rejection criteria as ASCAT (see above) apply, capable of flagging rainy WVCs. Furthermore, the OSI-SAF HY-2B and HY-2C wind products have been validated using buoys and triple collocation (Verhoef et al. 2024), showing wind quality well within the OSI-SAF product requirements (< 2 m/s wind component standard deviation, < 0.5 m/s speed bias). Against buoys, speed bias of -0.09/-0.06 m/s and u-/v-component standard deviations of 1.68/1.64 m/s (HY-2B) and 1.75/1.76 m/s (HY-2C) were found. Triple collocation gives scatterometer-perspective errors of 0.59/0.39 (HY-2B) and 0.81/0.65 (HY-2C) in the u-/v-components (Verhoef et al. 2024).

Availability:

HY-2B (launched 2018) and HY-2C (launched 2020) are operational and provide global coverage. However, currently only the most recent 3 days with a ~ 2.5 hour latency (rolling window) are automatically available in `12o-qc`. Data is available in both NetCDF and BUFR format.

Dataset acquisition:

OSI-SAF wind FTP server (`ftp.pro.knmi.nl`), directories `/scat/netcdf/hy2b` and `/scat/netcdf/hy2c`; registered OSI-SAF FTP credentials required. Only a 3-day rolling window is available.



Oceansat-3

Description:

10 m sea surface wind speed and direction wind-vector cells retrieved from the Ku-band scatterometer aboard ISRO's Oceansat-3 (EOS-06), reprocessed into the OSI-SAF wind-product format and redistributed via the same KNMI FTP server as HY-2B/HY-2C.

Characteristics:

same 25 km wind-vector-cell spacing and Ku-band, wide-swath pencil-beam geometry as HY-2B/HY-2C.

Data quality:

same `wvc_quality_flag`-based rejection criteria as ASCAT/HY-2B/HY-2C. Furthermore, the OSI-SAF Oceansat-3 wind products have been validated using buoys and triple collocation (Verhoef et al. 2024), showing wind quality well within the OSI-SAF product requirements (< 2 m/s wind component standard deviation, < 0.5 m/s speed bias). Against buoys, speed bias of -0.16 m/s and u/v component standard deviations of $1.72/1.68$ m/s were found (Verhoef et al. 2024). Triple collocation gives scatterometer-perspective errors of $0.68/0.54$ in the u/v-components (Verhoef et al. 2024).

Availability:

Oceansat-3 launched November 2022 and provides global coverage. Similarly to HY-2B and HY-2C, currently only the most recent 3 days with a ~ 2.5 hour latency (rolling window) are available in `12o-qc`. Data is available in both NetCDF and BUFR format.

Dataset acquisition:

OSI-SAF wind FTP server (`ftp.pro.knmi.nl`), directory `/scat/netcdf/oceansat3`; registered OSI-SAF FTP credentials required. Only a 3-day rolling window is available.

5.1.1.5 Radiometers

RSS microwave radiometers (AMSR2, GMI, SSMIS, WindSat)

Description:

Level-3 (L3), daily, global 10 m ocean surface wind speed products retrieved from passive microwave brightness temperatures, produced by Remote Sensing Systems (RSS) for six sensors: AMSR2 (GCOM-W1, JAXA), GMI (GPM, NASA/JAXA), SSMIS (DMSP F16/F17/F18; NASA), and WindSat (Coriolis, US Navy). WindSat additionally provides wind direction via its polarimetric channels; the other sensors are wind-speed-only.

Characteristics:

each sensor is distributed as a daily file already resampled to a common global 0.25° grid, with two passes (ascending/descending) per file and a per-cell measurement time. AMSR2 is published in NetCDF; GMI, SSMIS, and WindSat are distributed as gzipped RSS binary bytemaps. Wind speed is retrieved from a shared physical emissivity model (Meissner and Wentz 2012); cells under rain, land, or ice are already set to fill/NaN by RSS.

Data quality:

no per-cell quality bitmask is exposed; instead, invalid retrievals (rain-, land-, or ice-contaminated cells) are already flagged as fill values by RSS and dropped on read. Under rain-free conditions, RSS radiometer wind speed retrievals have a reported accuracy of ~ 1 m/s RMS against buoys (Meissner and Wentz 2012); WindSat wind direction retrieval accuracy (all-weather algorithm) ranges from $\sim 10^\circ$ in light rain to $\sim 30^\circ$ at the onset of heavy rain, with wind speed accuracy degrading from 2 m/s to 4 m/s over the same range (Meissner and Wentz 2009).

Availability:

AMSR2 (2012-present), GMI (2014-present), SSMIS F16 (2003-present), SSMIS F17 (2006-present), SSMIS F18 (2009-present), WindSat (2003-present); all provide global coverage via twice-daily (ascending/descending) overpasses.

Dataset acquisition:

Remote Sensing Systems public HTTPS archive (<https://data.remss.com/>), no account required; AMSR2 as NetCDF, GMI/SSMIS/WindSat as gzipped binary bytemaps.



5.1.1.6 Plane

SFMR

Dropsonde

5.1.2 Wind stress

5.2 Sea State

5.2.1 Total significant wave height

5.2.1.1 NWP

WAVEWATCH III

Description:

NOAA/NCEP's operational third-generation spectral wave model WAVEWATCH III (WW3), currently run as GFS-Wave and one-way coupled into the Global Forecast System (GFS) v16 via the NEMS/NUOPC coupling infrastructure. Used as an independent, physics-based wave-model reference, collocated via bilinear spatial and hyperbolic temporal interpolation. The current operational code version (v6.07, released by the WAVEWATCH III Development Group in 2019 as the model's first open-development GitHub release), run with the ST4 source-term physics package (wind input and dissipation) of (Ardhuin et al. 2010).

Characteristics:

global 0.16° regular latitude/longitude grid, with finer nested regional grids covering the Atlantic, East Pacific, and West Coast (all 0.16°), the Arctic (9 km), and the Southern Ocean (0.25°); the finest grid whose bounding box fully contains the query region is selected, falling back to the global grid otherwise. GFS-Wave initialises four times daily (00/06/12/18 UTC); 12o-qc reconstructs an hourly series from each cycle's short-range forecast leads. The model parameter significant height of combined wind waves and swell (HTSGW) is used as validation against SAR significant wave height.

Data quality:

no per-cell quality flag is provided in the GRIB2 output; land points are implicitly absent from the ocean model grid. The ST4 physics package used by the current operational configuration has been shown to outperform the older ST2 (Tolman–Chalikov) package used by earlier WAVEWATCH III code versions across most sea states, with the largest improvement in storm/extreme conditions add citation.

Availability:

From March 2021 (GFSv16/v6.07 implementation) – present, global coverage, published four times daily with hourly archived output between cycles.

Dataset acquisition:

NOAA Open Data on AWS, bucket `noaa-gfs-bdp-pds` (<https://noaa-gfs-bdp-pds.s3.amazonaws.com>), public and unsigned HTTPS/S3 access, no authentication required; GRIB2 format, parsed via the `eccodes/cfgrib` libraries.

5.2.1.2 C-band SAR

Sentinel-1 L2 OCN - OSW

Description:

The OSW component of the S1 L2_OCN product provides partition-based wave parameters, including a significant wave height `hs` for each identified wave partition, as well as a vignette-level total significant wave height [m], `total_hs`, and its standard deviation, `total_hs_standard_deviation`. The partition quantities describe individual wave systems resolved in the OSW wave spectrum, whereas `total_hs` represents an integrated sea-state estimate for the vignette. From IPF 3.9.0 onward the `total_hs` implementation is based on a variant of the deep-learning approach of Quach et al. (Quach et al. 2021). The OSW component is generated for Stripmap (SM) and Wave (WV) acquisition modes.

Characteristics:

in WV mode, acquisitions consist of isolated $\sim 20 \times 20$ km vignettes roughly 200 km apart along the orbit track, rather than a continuous swath, with one wave spectrum estimate per vignette.

In SM mode, the 80×170 km swath itself is tiled into a grid of 20×20 km OSW cells without gaps in between, unlike WV's sparse sampling. Each OSW cell contains its own independent wave spectrum/partition estimate.

Data quality:

The OSW product carries a total `quality_flag` (0 – 3), a per-partition `partition_quality_flag` (0 – 4), a `land_flag`, a per-partition `wave_direction_partition_ambiguity` flag, and a scalar `snr`, where for the total quality flag:

- 0: high quality
- 1: medium quality
- 2: low quality
- 3: bad quality

and, similarly but more finely graded, for the per-partition quality flag:

- 0: very good
- 1: good
- 2: medium
- 3: low
- 4: poor

The `wave_direction_partition_ambiguity` flag (0 or 1) indicates, per partition, whether the retrieved wave propagation direction is uniquely resolved or subject to the inherent 180° ambiguity of SAR-derived wave direction; as it concerns direction rather than the wave height itself, it is not used for filtering `total_hs`.

Data points are rejected when the total quality flag (`quality_flag`) is ≥ 2 , or when the land flag is set. However, the total quality flag currently only contains fill values as it does not seem to be populated by the operational processor.

Availability:

3 April 2014 – present. Unlike OWT's continuous-swath coverage, WV mode acquires sparse, isolated vignettes along the orbit track rather than continuous global coverage. Note that the total significant wave height, `total_hs`, is only available since 2022 (Johnsen et al. 2022). In addition, `total_hs` was unavailable for `ipf_version` 3.8.0, after which it was reactivated from `ipf_version` 3.9.0 onwards. The significant wave height partition, `hs`, was not affected by these changes.

Dataset acquisition:

Copernicus Data Space Ecosystem (<https://dataspace.copernicus.eu/>); OData catalogue/download API at <https://catalogue.dataspace.copernicus.eu/odata/v1/Products>; free registration required.

5.2.1.3 In-Situ Observations**ECMWF MARS archive****Description:****Characteristics:****Data quality:****Availability:****Dataset acquisition:**

Copernicus Marine

Description:

Dataset ID INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030, which contains in-situ observations. Relevant platforms reporting significant wave height are buoys, moorings, and tidal gauges.

Characteristics:

point observations at native platform sampling rate; default collocation, which is user adjustable, is ± 30 min / 5 km aggregation radius.

Data quality:

Copernicus Marine in situ TAC aggregates operational oceanography data and metadata from multiple providers, ingested with varying confidence depending on the data source (Copernicus In Situ Tac Data Management Team 2025). Bad flags are preserved and an additional quality control is performed that may result in a QC flag degradation (never an upgrade). For waves, this additional quality control procedure applies 7 different tests to wave parameters as described in the Copernicus Marine In Situ TAC quality control procedure (2025) (Copernicus In Situ Tac Data Management Team 2025).

The quality flag ranges from 0 – 9:

- 0: No QC was performed
- 1: Good data
- 2: Probably good data
- 3: Bad data although potentially correctable
- 4: Bad data
- 5: Value changed
- 6: Value below detection/quantification
- 7: Nominal value (e.g. reported rather than observed value like instrument target depth)
- 8: Interpolated value
- 9: Missing value

Accepted QC flag codes are {1, 2, 5, 7, 8}. A significant part of the data available on Copernicus Marine In situ lacks a QC and is thus dropped before validation.

Accuracy of the measurements provided by Copernicus Marine is sensor dependent, but all sensors have an uncertainty of maximum 1% or 0.1 m (Wehde et al. 2025). The estimated accuracy due to statistical variability is < 5% for SWH (Wehde et al. 2025).

Availability:

Platform dependent, some platforms provide a long history. Not all platforms have both WSPD and WDIR available. Geographical coverage can be seen in Figure Fig. 5.3 for the first half of August 2026. Moorings and tidal gauges have fixed locations, while the buoys drift.

Dataset acquisition:

Copernicus Marine in-situ dataset (https://data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/), download via the copernicusmarine Python client (free registration).

5.2.1.4 Sea-ice maps

TBD

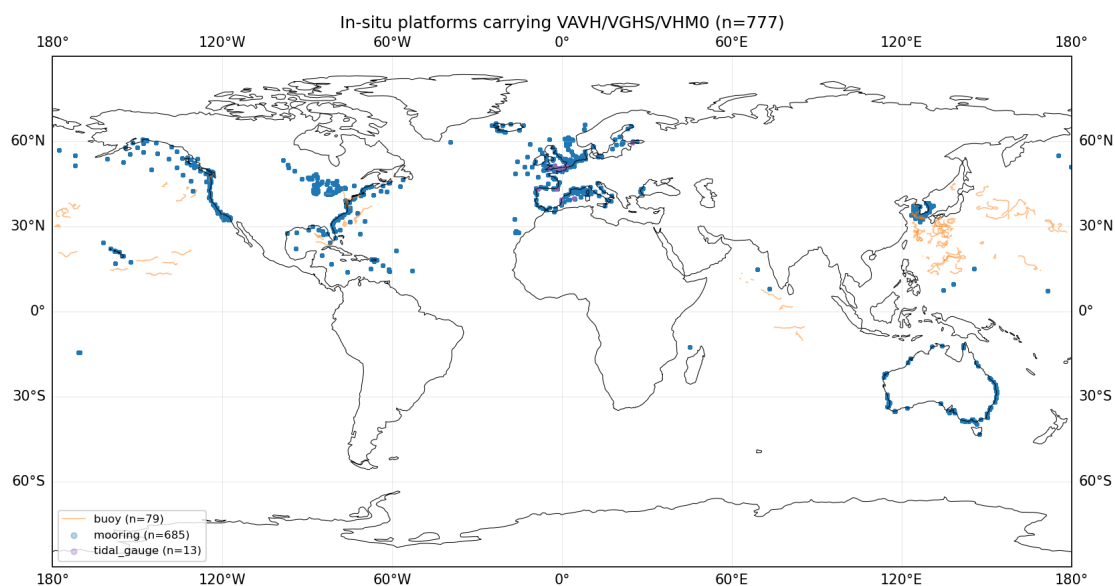


Fig. 5.3: Coverage of Copernicus Marine in-situ platforms that provided significant wave height during the first half of August 2026.

5.2.1.5 Altimeters

SWH altimeter observations are divided into a reprocessed version that currently has 1 Hz altimeter data from 03-08-1991 - 31-12-2023 (Charles and Ollivier, n.d.) and a near-real time (NRT) version that contains 1 Hz sampled altimeter data from 28-12-2023 - present and 5 Hz sampled altimeter data from 09-03-2026 - present (Philip et al., n.d.).

Global Ocean L3 Significant Wave Height from Reprocessed Satellite Measurements

Description:

This reprocessed dataset provides significant wave height data derived from satellite altimeter data including the Topex, ERS-1/2, Jason-1/2/3, Envisat, Saral/AltiKa, Cryosat-2, and Sentinel-3A/3B/6A missions as well as altimeter L3 V4 data from the ESA Climate Change Initiative (CCI) Sea State project. The along-track measurements from the different missions are combined into a single daily file, thus resulting in a unified dataset combining all available missions. The dataset includes distance to coast, sea floor depth below mean sea level, sea surface wave significant height (regular, adjusted, denoised) and its uncertainty (Charles and Ollivier, n.d.).

Characteristics:

The 1 Hz sampling corresponds to an along track resolution of 7 km, resulting in a default aggregation window of 7 km and 180 min in the 12o-qc tool. Jason-2 serves as a calibration anchor and CCIv4 improves cross-calibration across all missions. The denoised SWH is used as validation dataset. Denoising means cross-mission bias correction and Empirical Mode Decomposition (EMD) denoising following Quilfen and Chapron (2019) (Quilfen and Chapron 2021).

For Topex, Jason-1/2/3, and Sentinel 6-A, the coverage is maximum 66°S/66°N in latitude; for SARAL/AltiKa 81.5°S/81.5°N; for Cryosat-2 88°S/88°N; for ERS-1/2, Envisat, and Sentinel-3A/3B 90°S/90°N (Charles and Ollivier, n.d.).

Data Quality:

A strict QC filter is applied before creation of the dataset, only allowing `swh_quality_level = 3` (good data) to be included. No QC filtering is needed in the 12o-qc tool and no quality flags themselves are provided in the dataset.

Availability:

03-08-1991 - 31-12-2023; 1 Hz sampling frequency.



Dataset acquisition:

Copernicus Marine satellite altimetry dataset (https://data.marine.copernicus.eu/product/WAVE_GLO_PHY_SWH_L3_MY_014_005/), download via the `copernicusmarine` Python client (free registration).

Global Ocean L3 Significant Wave Height from NRT Satellite Measurements**Description:**

The product consists of an inter-calibrated, denoised multi-mission altimeter-derived along-track SWH dataset. It combines 1 Hz sampling data from CFOSAT, CryoSat-2, HY-2B/2C, Jason-3, Saral/AltiKa, Sentinel-3A/3B/6A, and SWOT nadir producing sea surface wave significant height (filtered and unfiltered) as well as sea surface wind speed. The 5 Hz sampling contains data from CryoSat-2, Jason-3, Saral/AltiKa, and Sentinel-3A/3B/6A and produces sea surface wave significant height (filtered and unfiltered).

Characteristics:

The data from the different missions are homogenized with respect to a reference mission (Jason-3 until April 2022, Sentinel-6A afterwards). The 1 Hz sampling corresponds to an along track resolution of 7 km, resulting in a default aggregation window of 7 km and 180 min. The 5 Hz sampling has an along track resolution of 1.4 km, resulting in a default aggregation window of 1.4 km and 180 min in the `l2o-qc` tool.

For Jason-3, Sentinel-6A and Hai Yang-2C, the coverage is maximum 66°S/66°N in latitude; for Sentinel-3A, Sentinel-3B and AltiKa, 81.5°S/81.5°N; for Cryosat-2, 88°S/88°N; for HaiYang-2B, 81°S/81°N; for CFOSAT, 83°S/83°N and for SWOT, 78°S/78°N. Note that for Sentinel-3A and B, the altimeter wave measurements are deduced from both SAR measurements.

Data Quality:

Prior to inclusion in the dataset, data is selected based on a combination of various criteria such as quality flags and parameter thresholds, including surface flag, presence of ice, sigma0 (mission dependent), SWH (min: 0 m; max: 30 m), wind speed (min: 0 m/s; max: 30 m/s), sigma0 standard deviation (min: 0 dB; max 3 dB), shoreline distance (≥ 0) (Philip et al., n.d.). Only data points passing these criteria are included in the dataset. No QC filtering is needed in the `l2o-qc` tool and no quality flags themselves are provided in the dataset.

Availability:

1 Hz sampling rate: late 2023 - present; note that the NRT 1 Hz dataset is used from 01-01-2024 onward to replace the reprocessed one that ends 31-12-2013. 5 Hz sampling rate: 09-03-2026 - present. Availability delay is 3 hours for all satellites other than HY-2B/2C, which are provided with a delay of 7 hours.

Dataset acquisition:

Copernicus Marine satellite altimetry dataset (https://data.marine.copernicus.eu/product/WAVE_GLO_PHY_SWH_L3_NRT_014_001/), download via the `copernicusmarine` Python client (free registration).

5.2.2 Mean period wind-sea

5.2.3 First moment wave period

5.2.4 Second moment wave period

5.2.5 Wave height swell dominant system

5.2.6 Wave height swell secondary system

5.2.7 Significant wave height wind-sea

5.2.8 Mean period wind-sea

5.2.9 Wave direction

5.2.10 Wave directional energy spectrum

5.3 Line-of-sight currents

5.3.1 LoS wave Doppler

5.3.2 LoS surface current component

5.3.2.1 NWP

HYCOM

Description:

The HYbrid Coordinate Ocean Model (HYCOM) is a data-assimilative, eddy-resolving global ocean general circulation model producing three-dimensional fields of ocean state (Chassignet et al. 2009). Its vertical coordinate is hybrid, blending isopycnal (density-following) layers in the stratified open ocean, terrain-following layers in shallow coastal regions, and fixed pressure-level coordinates in the surface mixed layer and/or unstratified open seas. This toolbox uses HYCOM's surface-level ($z = 0$) horizontal current velocity, `water_u/water_v`, drawn from two HYCOM configurations: Global Ocean Forecasting System (GOFS) 3.1 Analysis (GLBy0.08/expt_93.0; 04-12-2018 - 10-08-2024) (Metzger et al. 2017), and its higher-resolution successor ESPC-D-V02 (10-08-2024 onwards), the ocean component of the Navy's Earth System Prediction Capability (Navy-ESPC) (Barton et al. 2021). The eastward/northward current vector at each grid cell is projected onto the local SAR LoS (radial) direction for direct comparison against radial surface-velocity measurements, using the same heading-based projection of the SAR satellite applied throughout this validation component.

Characteristics:

GOFS 3.1 Analysis runs at a $1/12^\circ$ global HYCOM resolution (~ 9 km at equator, ~ 7 km at mid-latitudes, ~ 3.5 km at North Pole); ESPC-D-V02's can run at a higher, $1/25^\circ$ ($\sim 3\text{--}4$ km at mid-latitudes), resolution (Barton et al. 2021). However, the version used by 12o-qc is regridded onto the same grid as GOFS 3.1. HYCOM consists of 3-hourly instantaneous analyses with 40 standard z -levels of which 12o-qc selects the surface level (depth = 0.0). GLBy0.08/expt_93.0 publishes one combined `water_u+water_v` NetCDF/OPeNDAP dataset per calendar year; ESPC-D-V02 publishes one continuous OPeNDAP dataset per component spanning its whole coverage range.

Data Quality:

As a data-assimilative reanalysis/forecast product rather than a direct retrieval, HYCOM carries no per-cell observational quality flag. Instead its data quality depends on the underlying model, its initial state, and data assimilation checks. For ESPC-D-V02, all assimilated data go through a quality control system checking for gross error and assigning a likelihood of error based on comparison to ocean analyses (instrument error and representativeness error) (Barton et al. 2021).

Availability:

GOFS 3.1 Analysis (expt_93.0): 04-12-2018 - 04-09-2024 (used by 12o-qc until 10-08-2024 12:00 UT). ESPC-D-V02: 10-08-2024 - present.

Dataset acquisition:

HYCOM public THREDDS Data Server (<https://tds.hycom.org/thredds/>), OPeNDAP endpoints [https://tds.hycom.org/thredds/dodsC/ESPC-D-V02/u3z \(+v3z\)](https://tds.hycom.org/thredds/dodsC/ESPC-D-V02/u3z (+v3z)) and https://tds.hycom.org/thredds/dodsC/GLBy0.08/expt_93.0/uv3z/{year}; openly accessible, no registration required. See also <https://www.hycom.org/dataserver>.

5.3.2.2 C-band SAR**Sentinel-1 L2 OCN - RVL****Description:**

The RVL (Radial Velocity) component of the Sentinel-1 L2_OCN product provides an estimate of the total Doppler frequency [Hz] and the corresponding radial LoS surface velocity [m/s], `rvlRadVel`, obtained by fitting (least square minimization) the antenna model to the observed azimuth Doppler spectrum of the Level-1 SLC image. The estimated Doppler frequency is then corrected for antenna mis-pointing (as a function of elevation angle) and the attitude/orbit Doppler error (Engen and Johnsen 2011). Other relevant parameters in the RVL component include standard deviation, `rvlRadVelStd`, and the acquisition geometry variables `rvlHeading` (along-track heading) and `rvlIncidenceAngle`. The RVL component is generated for all four Sentinel-1 imaging modes: WV, SM, IW, and EW (Engen and Johnsen 2011).

Characteristics:

The spatial coverage of the RVL product equals that of the corresponding mode. In WV mode, RVL is thus computed on the same isolated $\sim 20 \times 20$ km vignette used by the rest of that mode's OCN product, at roughly 200 km spacing along the orbit track. In SM/IW/EW mode, RVL is produced on a grid $\sim 1 \times 1$ km similar to the OWI products. To compare an in-situ or model current vector (EWCT, NSCT) against this scalar radial measurement, the vector is projected onto the SAR line-of-sight in the `l2o-qc` toolbox using

$$v_{\text{projection}} = \text{EWCT} \cdot \cos(\text{rvlHeading} - 90^\circ) + \text{NSCT} \cdot \sin(\text{rvlHeading} - 90^\circ),$$

where `rvlHeading` is stored in mathematical convention (counter-clockwise from East) and the -90° term converts the along-track heading to the (right-looking) range direction.

Data Quality:

In the RVL component, `rvlLandFlag` is automatically set to 1 where a cell's land coverage exceeds 10% (Engen and Johnsen 2011). `l2o-qc` then masks both `rvlRadVel` and `rvlRadVelStd` as NaN'd at flagged cells. No separate per-cell QC-code flag exists for RVL beyond the land flag. Note that the operational `rvlRadVel` is known to carry substantial platform-induced biases, up to ~ 2 m/s, from residual antenna mis-pointing and instrument effects (Martin et al. 2022). This is currently not compensated for in the `l2o-qc` toolbox.

Availability:

3 April 2014 – present, across all four imaging modes (WV, SM, IW, EW).

Dataset acquisition:

Copernicus Data Space Ecosystem (<https://dataspace.copernicus.eu/>); OData catalogue/download API at <https://catalogue.dataspace.copernicus.eu/odata/v1/Products>; free registration required.

5.3.2.3 HF radars**Copernicus Marine - global radar coverage****Description:**

Gridded, total-vector surface-current field (EWCT and NSCT) from Copernicus Marine. Contributors are mainly European national HF-radar networks together with NOAA (further described in Section [NOAA - US radar coverage](#)); every European region is processed through a single, centrally-coordinated pipeline, the EU HFR Node (Lorente et al. 2022). Additional variables are per-cell standard deviations EWCS/NSCS, a geometric-dilution-of-precision field GDOP, and several QC flags including an overall QCflag.

Characteristics:

25 named regions, spanning the Mediterranean, Atlantic, North Sea, and Norwegian coasts (e.g. Galicia, Gibraltar, DeltaEbro, COSYNA, Skagerrak, Finnmark, Vestlandet) as well as the five US coastal areas also served by NOAA (Figures Fig. 5.4 and Fig. 5.5), each operated by a different national institution (Table Table 5.1) but converging on the same EU HFR Node distribution pipeline for the European regions. Near-real-time (latest-radar-total) coverage exists for 19 of the 25 regions; the remaining seven, ARPAS, COSYNA, Finnmark, US-Alaska, US-Hawaii, and WHub, are available only through the monthly/delayed-mode feed. Default layer-vs-layer collocation against the SAR grid is ± 30 min with a 6 km cell-averaging aggregation window and equal spatial weighting. This is adjustable through the .yaml recipe file. Native grid resolution and cadence vary by region, where most regions contain hourly data. Exceptions are COSYNA (20 min) and NAdr (30 min). All available resolution are listed in Table Table 5.1.

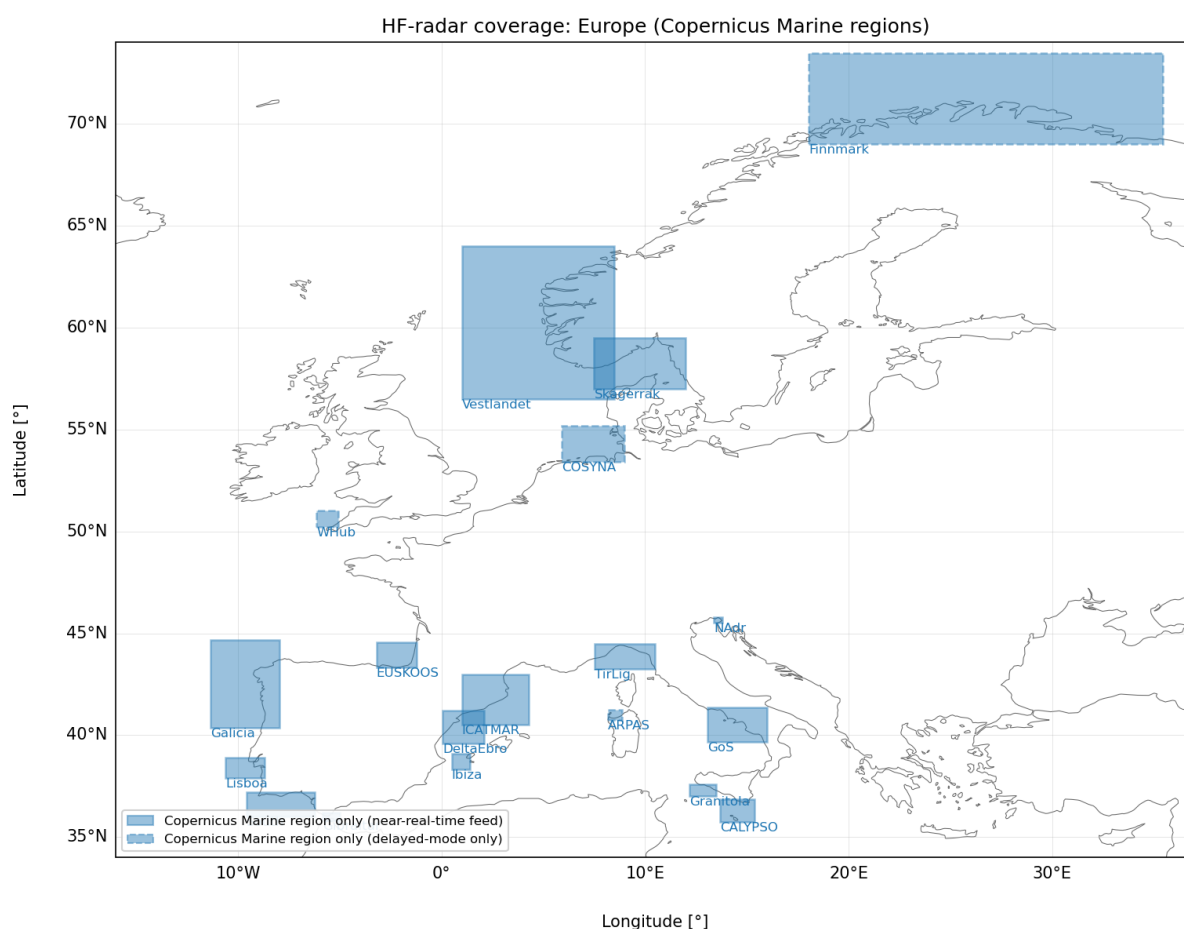


Fig. 5.4: Bounding boxes of regions available in Copernicus Marine HF-radar dataset across Europe and the Mediterranean. Regions that are no longer available in real-time (delayed-mode only) have dashed lines.

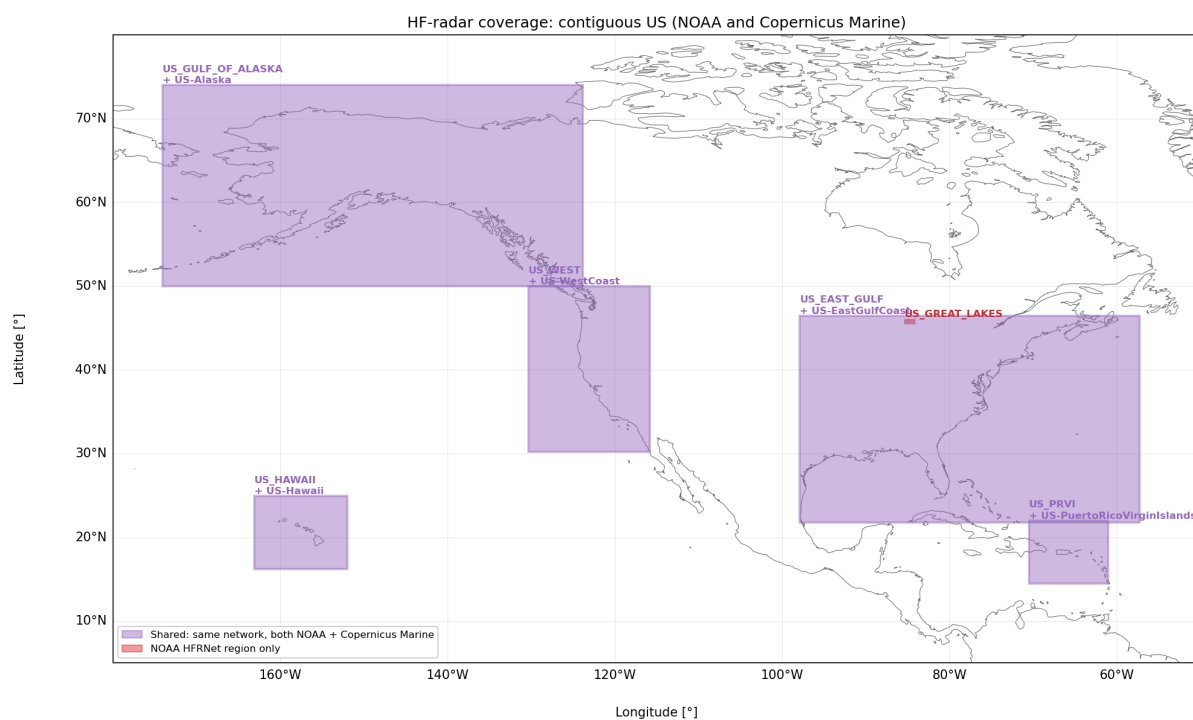


Fig. 5.5: Bounding boxes of the 5 Copernicus Marine HF-radar regions and 6 NOAA regions. Regions with that have overlapping coverage from both are purple, while the region only provided by NOAA is red.

Table 5.1: Copernicus Marine HF-radar region grid resolution and upstream QC thresholds, read directly from each region's monthly-radar-total metadata and EU HFR Node dataset attributes. Regions without footnote have self descriptive place names.

Region	Spatial res.	Temporal res.	Max GDOP	Max speed
ARPAS	2 km	60 min	2.0	1.2 m/s
CALYPSO	3 km	60 min	20.0	1.0 m/s
COSYNA	2 km	20 min	2.0	1.2 m/s
DeltaEbro	3 km	60 min	2.0	1.2 m/s
EUSKOOS	5 km	60 min	2.0	1.2 m/s
Finnmark	5 km	60 min	2.0	1.2 m/s
Galicla	6 km	60 min	2.0	1.2 m/s
Gibraltar	1 km	60 min	2.0	2.5 m/s
GoS	2 km	60 min	2.0	1.2 m/s
Granitola	1.5 km	60 min	2.0	1.2 m/s
ICATMAR	3 km	60 min	2.0	1.7 m/s
Ibiza	3 km	60 min	2.83	0.7 m/s
Lisboa	1.4 km	60 min	2.0	1.2 m/s
NAdr	1.5 km	30 min	2.0	1.0 m/s
PLOCAN	0.75 km	60 min	2.0	1.2 m/s
Skagerrak	3 km	60 min	2.0	1.2 m/s
South	1.5 km	60 min	2.0	1.2 m/s
TirLig	2 km	60 min	2.0	1.2 m/s
US-Alaska	6 km	60 min	2.0	5.0 m/s
US-EastGulfCoast	6 km	60 min	2.0	5.0 m/s
US-Hawaii	6 km	60 min	2.0	5.0 m/s
US-PuertoRicoVirginIslands	6 km	60 min	2.0	5.0 m/s
US-WestCoast	6 km	60 min	2.0	5.0 m/s
Vestlandet	5 km	60 min	2.0	1.2 m/s
WHub	1 km	60 min	2.0	1.2 m/s

Northern Sardinia, Tyrrhenian Sea. Malta Channel, Western Mediterranean. German Bight, Germany. Ebro River Delta, SE Spain. Basque Country coast, Spain. Gulf of Salerno, Tyrrhenian Sea, Italy. Catalan coast, Spain. Gulf of Trieste, Northern Adriatic Sea. North-east coast of Gran Canaria, Spain. Portuguese/Spanish south coast, 4 sites. North-western Tyrrhenian and Ligurian Sea, Italy. Wave Hub test site, off North Cornwall, UK.

Data quality:

Cells are kept only where the overall QCflag, for value meaning see section *Copernicus Marine*, is equal to {1, 2, 5, 7, 8} (Copernicus In Situ Tac Data Management Team 2025)). Per-parameter flags (CSPD_QC, DDNS_QC, GDOP_QC, VART_QC, POSITION_QC) are retained but not used for filtering. QC flagging uses a maximum Geometric Dilution of Precision (GDOP) threshold and a maximum velocity for each radar, which are listed in Table [Table 5.1](#). Any data point exceeding those values gets flagged.

Data quality:

Cells are kept only where the overall QCflag, for value meaning see section *Copernicus Marine*, is equal to {1, 2, 5, 7, 8} (Copernicus In Situ Tac Data Management Team 2025)). Per-parameter flags (CSPD_QC, DDNS_QC, GDOP_QC, VART_QC, POSITION_QC) are retained but not used for filtering. Five upstream tests, defined by the EU HFR Node's own QC recommendations (Corngati et al. 2024, Table 2), feed into the overall QCflag: DDNS_QC (Data Density Threshold) checks that at least a minimum number of three radial velocities contributed to the total-vector solution; CSPD_QC (Velocity Threshold) checks the total current speed against a region-specific maximum (see Table [Table 5.1](#); VART_QC (Variance Threshold or Temporal Derivative) checks the solution's own temporal variance (only relevant for beam-forming systems); GDOP_QC (Geometric Dilution of Precision (GDOP) Threshold) checks the GDOP against a region-specific maximum (see Table [Table 5.1](#)); and POSITION_QC independently checks the solution's own reported latitude/longitude, unrelated to the four velocity-based tests. Any data point failing at least one of the QC tests gets flagged.

Availability:

Region-dependent. The near-real-time feed (last 30 days) is available for 19 of the 25 regions; the monthly feed (from up to 01-01-2020 - present (< 30 day latency)) and historic archive (typically large single file per radar) are available for all 25 regions.

Dataset acquisition:

Copernicus Marine near-real-time in-situ dataset (https://data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/description; latest (last 30 days) and monthly (from 01-01-2020) available), radar-total dataset_part; delayed-mode currents dataset (https://data.marine.copernicus.eu/product/INSITU_GLO_PHY_UV_DISCRETE_MY_013_044/description), radar-total product; both via the copernicusmarine Python client (free registration).

NOAA - US radar coverage

Description:

Gridded, hourly total-vector surface-current fields (water_u and water_v) from the US Integrated Ocean Observing System (IOOS) National HF Radar Network (HFRNet), combined from individual radial-current sites and distributed by NOAA. The dataset is distributed via two different backends, ERDDAP griddap endpoint covering a rolling ~90-day real-time window, and an NCEI THREDDS archive spanning 2006-present (latency of a few weeks). The 12o-qc toolbox tries the ERDDAP backend first, falling back to THREDDS for older dates the rolling window no longer covers. In case of no availability via NOAA, the Copernicus Marine HF radar database, as described in Section *Copernicus Marine - global radar coverage*, is tried.

Characteristics:

A regular (time, lat, lon) grid at region-dependent resolutions of 0.5, 1, 2, or 6 km, at hourly cadence, across six regions: US West Coast, US East Coast/Gulf, Great Lakes, Hawaii, Puerto Rico/US Virgin Islands, and the Gulf of Alaska. See Table [Table 5.2](#) for available resolutions per region and retrieval mechanism. The locations of these regions can be seen in Figure [Fig. 5.5](#) shows their bounding boxes).

Table 5.2: NOAA HF-radar resolution availability per region, by backend.

Region	ERDDAP	THREDDS	Default
US_WEST	0.5, 1, 2, 6 km	0.5, 1, 2, 6 km	6 km
US_GREAT_LAKES	none	0.5, 1, 2, 6 km	6 km
US_EAST_GULF	1, 2, 6 km	1, 2, 6 km	6 km
US_HAWAII	1 km only	1, 2, 6 km	1 km
US_PRVI	2, 6 km	2, 6 km	6 km
US_GULF_OF_ALASKA	none	2, 6 km	6 km

West Coast

East Coast/Gulf

Puerto Rico/US Virgin Islands

Resolutions the ERDDAP backend supports per its own catalogue definition, not always live accessible as individual region/resolution datasets on NOAA's rolling real-time feed can transiently 404.

Data quality:

NOAA's HF-radar product carries no discrete per-cell QC flag on the delivered total-vector grid. Individual radials are filtered upstream before total-vector computation including among others a maximum-magnitude threshold, and a land mask (Otero 2021). Therefore, any value present in the retrieved data has already passed filtering.

Availability:

ERDDAP: rolling ~90-day window ending near real time. THREDDS archive: 2006 – present (latency of a few weeks). See the per region availability in Table Table 5.2. Together these archives cover a span of 2006 – present for most of the regions.

Dataset acquisition:

NOAA CoastWatch ERDDAP griddap server (<https://coastwatch.pfeg.noaa.gov/erddap/griddap/>); NOAA NCEI THREDDS HF-radar archive (<https://www.ncei.noaa.gov/thredds-ocean/>); both public, no registration required.

5.3.2.4 In-Situ Observations (Copernicus Marine)

Description:

Dataset ID INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030, which contains in-situ observations. Relevant platforms reporting surface currents are buoys, moorings, drifters, and ferryboxes.

Characteristics:

point observations at native platform sampling rate; default collocation, which is user adjustable, is ± 30 min / 5 km aggregation radius.

Data quality:

Real time quality control procedure subjects the currents (meta)data to 21 tests as described in the Copernicus Marine In Situ TAC quality control procedure (2025) (Copernicus In Situ Tac Data Management Team 2025). The resulting QC flag for current parameters, ranging from 0 – 9, is the same as the one described in Section *Copernicus Marine*. Accepted QC flag codes are again {1, 2, 5, 7, 8}.

Accuracy of current measurements are expected to be 0.5 - 1 cm/s (Wehde et al. 2025).

Availability:

Platform dependent, the Copernicus Marine dataset provides data from 1 January 2020 - present. Not all platforms measure surface currents. The geographical coverage of the platforms that do can be seen in Figure Fig. 5.6 for the first half of August 2026. Moorings have fixed locations, while the other platforms drift with the currents.

Dataset acquisition:

Copernicus Marine in-situ dataset (https://data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/), download via the copernicusmarine Python client (free registration).



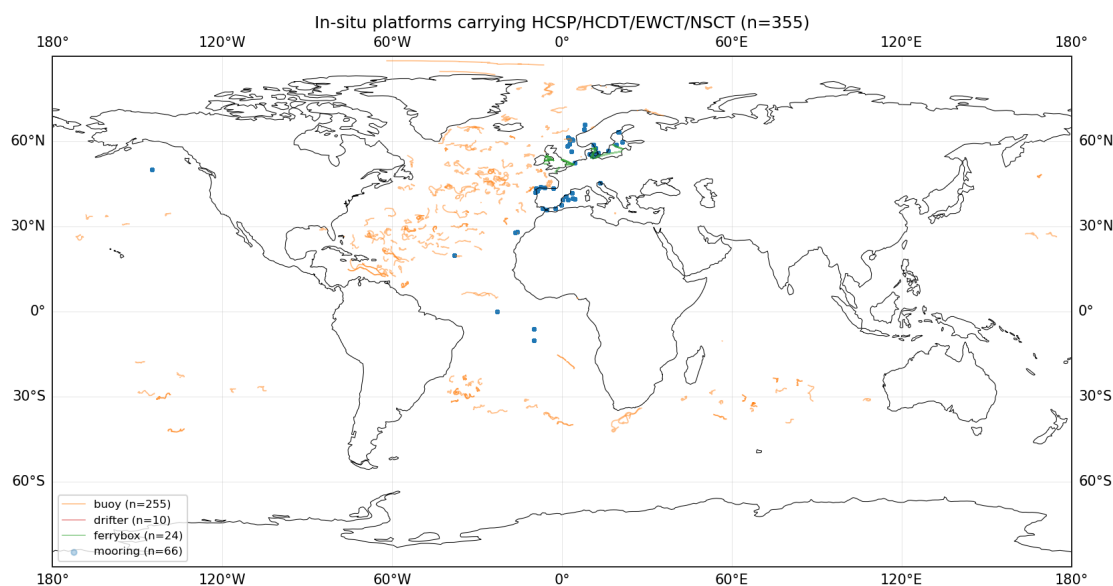


Fig. 5.6: Coverage of Copernicus Marine in-situ platforms that provided sea surface currents during the first half of August 2026.

5.3.2.5 Altimeters

5.3.2.6 IWRAP

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ALGORITHMS DESCRIPTION

The NOC calibration method and four validation methods are described in the following pages.

6.1 Selected Calibration methods

6.1.1 ROSE-L L1 SLC products

6.1.1.1 Short description

6.1.1.2 Description of inputs

6.1.1.3 Detailed methodology

6.1.1.4 Performance metrics

6.1.1.5 Expected output

6.1.2 Numerical Weather Prediction Ocean Calibration of SAR σ_0

The aim of the Numerical Weather Prediction Ocean Calibration (NOC) (*Verspeek et al. (2012)*) is to provide an additional calibration step to the SAR averaged normalized radar cross section (σ_0) values used to retrieve the wind vector. These values are generally averaged on a grid of a few kilometres (approximately $1 - 2 \times 1 - 2 \text{ km}^2$).

This calibration procedure is not applicable to sea-state parameters, since these are derived spectrally. Therefore, an absolute σ_0 calibration is not necessary before retrieval. Instead, inter-calibration among different sea-state parameter sources can be useful for comparison and validation; this can be achieved with triple collocation (TC). Similarly, no additional SAR-specific calibration procedure is recommended for line-of-sight (LoS) current retrieval beyond the harmonized Doppler calibration already part of the 12o processor.

Inputs

- SAR σ_0 values used to retrieve the wind-vector field.
- Collocated wind vector from numerical weather prediction (NWP) models.

Procedure

1. Set up approximately 10^7 NWP-wind/SAR- σ_0 collocations, to ensure that wind direction is sampled uniformly and that the wind-speed range is adequately represented. Correct σ_0 for Faraday rotation, which arises from ionospheric propagation effects at L-band and lower frequencies. Obtaining this many collocations requires 1–2 months of acquisitions, depending on revisit frequency and coverage.
2. Bin collocations by incidence angle, wind direction, and wind speed, with bin sizes of 1° , 1° , and 1 m s^{-1} , respectively.
3. Compute the expected (simulated) σ_0 values using the appropriate geophysical model function (GMF):

$$\sigma_0^{W,\text{sim}} = \text{GMF}(\vec{u}^{\text{NWP}}, \theta, f_c, \text{pol})$$

Here, $\vec{u}^{\text{NWP}}$ is the NWP wind vector, and θ , f_c , and pol are the SAR incidence angle, carrier frequency of the transmitted signal, and polarization channel, respectively. The intended model is an L-band GMF (LMOD), analogous to the CMOD7 GMF used for C-band SAR satellites such as Sentinel-1 (*Stoffelen et al. (2017)*). LMOD specification needs to be determined.



4. Obtain the final calibration-coefficient profile by weight-averaging the ratios of simulated and measured σ_0^W values over wind direction and then wind speed:

$$\alpha(\theta) = \left\langle \left\langle \left\langle \frac{\sigma_0^{W,\text{sim},ijk}(\theta)}{\sigma_0^{W,m,ij}(\theta)} \right\rangle_k \right\rangle_j \right\rangle_i.$$

Here, α is the calibration correction factor; k is the index of occurrences in the $\{i, j\}$ bin; and j and i are the wind-direction and wind-speed indices, respectively. The angle-bracket operator denotes the ensemble average.

Iterate the procedure until $\alpha(\theta)$ is close to 1, according to the selected threshold:

$$\Delta_\alpha = \left| \frac{\sigma_0^{W,\text{sim}}}{\sigma_0^{W,c}} - 1 \right| \leq \Delta_\alpha^{\text{th}}.$$

The corrected σ_0^W values are:

$$\sigma_0^{W,c}(\theta) = \alpha(\theta) \sigma_0^{W,m}(\theta).$$

In logarithmic space:

$$\sigma_{0,\text{dB}}^{W,c}(\theta) = \alpha_{\text{dB}}(\theta) + \sigma_{0,\text{dB}}^{W,m}(\theta).$$

Metrics

$$\Delta_\alpha = \left| \frac{\sigma_0^{W,\text{sim}}}{\sigma_0^{W,c}} - 1 \right|.$$

Outputs

- Calibration coefficients: $\alpha(\theta)$ (equivalently $\alpha_{\text{dB}}(\theta)$).
- Calibration residuals: Δ_α .

6.1.3 GMF tuning

6.1.3.1 Short description

6.1.3.2 Description of inputs

6.1.3.3 Detailed methodology

6.1.3.4 Performance metrics

6.1.3.5 Expected output

6.1.4 MTF tuning

6.1.4.1 Short description

6.1.4.2 Description of inputs

6.1.4.3 Detailed methodology

6.1.4.4 Performance metrics

6.1.4.5 Expected output

6.1.5 Summary

References

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6.2 Validation

6.2.1 Spatial Variance

Spatial Variance is a method used to estimate the representativeness error r^2 , which is a key component for triple collocation (TC). This procedure estimates the spatial variance for a given geophysical parameter. Let s_1 and s_2 be two different sources characterized by different spatial resolutions, where $V_{s_1}(n)$ and $V_{s_2}(n)$ represent their spatial variances at spatial lag n . Given $n_{\max}$, the maximum lag, the representativeness error of s_1 (the coarsest system) compared to s_2 can be estimated as:

$$r^2 = \sum_{n=0}^{n_{\max}} [V_{s_2}(n) - V_{s_1}(n)]. \quad (6.1)$$

Let $\tilde{n} < n_{\max}$ be the spatial-lag threshold above which both systems capture the same signal variability. The residual contribution then vanishes:

$$V_{s_2}(n) - V_{s_1}(n) \rightarrow 0 \quad \forall n > \tilde{n}.$$

Purpose

This method (Vogelzang *et al.* (2015)) computes wind variances as a function of scale. Unlike the spectral-integration method, it operates in the spatial domain rather than the frequency domain. It is more tolerant to missing samples and provides a more intuitive interpretation. The representativeness error is the difference between the system 2 (e.g. satellite) spatial variance and that of system 3 (e.g. model). This value depends on the selected scale. Spatial variances increase monotonically with scale, and their difference also usually increases up to a certain scale.

Inputs

- Vector samples of the required geophysical parameter.

Procedure

1. Estimate the first-order moments for both components of the geophysical parameter of interest:

$$M_{w,i}(n) = \frac{1}{n+1} \sum_{j=0}^n w_{i+j}. \quad (6.2)$$

Here, w is the component of the geophysical parameter of interest, n is the lag size, and i is an ancillary index identifying the starting element.

2. Estimate the second-order moments:

$$M_{ww,i}(n) = \frac{1}{n+1} \sum_{j=0}^n w_{i+j}^2. \quad (6.3)$$

The subscript w denotes the first-order moment, while ww denotes the second-order moment.

3. Estimate the spatial variance:

$$V_{w,i}(n) = \langle M_{ww,i}(n) - M_{w,i}^2(n) \rangle. \quad (6.4)$$

Here, angle brackets denote the expected-value operator.

4. Average $V_{w,i}$ to reduce uncertainty:

$$V_w(n) = \frac{1}{M} \sum_{i=0}^M V_{w,i}(n), \quad M = N - n. \quad (6.5)$$

⁵ <https://doi.org/10.1109/JSTARS.2017.2681806>

Outputs

- $V_{u,v}(n)$.

6.2.2 Triple Collocation

Purpose

Triple collocation (TC) was conceived by Stoffelen (1998) as a tool for inter-calibration and individual error assessment of three different collocated sea-surface wind datasets (*Stoffelen (1998)*). TC is generic and can be applied to any geophysical variable and data product, provided that errors are additive, linear calibration is sufficient, measurement errors are uncorrelated with each other, and measurement errors are constant over the range of measured values.

SAR-derived wind products are recommended to be validated against buoy wind measurements, scatterometer-derived winds, and NWP winds. Collocations between SAR and scatterometers strictly depend on the Local Time Ascending Node (LTAN). TC analysis needs a reliable estimate of the representativeness error r^2 ; see *Spatial Variance* for its estimation.

Inputs

One or more triplets including the SAR-derived parameter of interest. For wind, the following sources are relevant:

- SAR-derived wind vector.
- Scatterometer-derived wind vector.
- Buoy wind vector.
- NWP wind vector.

When available, buoy and scatterometer measurements are preferred to NWP winds, due to their higher spatial resolution.

For sea-state parameters such as significant wave height (SWH), relevant validation sources are:

- SAR-derived SWH.
- Altimeter-derived SWH.
- Buoy SWH measurement.
- SWH from an NWP model.

For LoS surface currents, relevant validation sources are:

- SAR-derived LoS surface currents.
- HF-radar total surface-current vector.
- Buoy total surface-current vector.
- LoS surface currents from an NWP model.

Procedure

Given three measurement systems, $i = 1, 2, 3$, representing, for example, in-situ, satellite, and model estimates, the measurements and measurement errors are approximated by the following linear expression (*Vogelzang and Stoffelen (2021)*; *Vogelzang and Stoffelen (2022)*):

$$s_i = a_i(t + \delta_i) + b_i. \quad (6.6)$$

Here, t is the common quantity, i.e. the true value of the parameter at the scales commonly resolved by all three data sources. a_i and b_i are the scaling and offset (bias) calibration coefficients, respectively, and δ_i is the random measurement error. For a vector quantity such as wind, s can represent either the zonal (u) or meridional (v) component. Then:

1. Collocate triplets in space and time. For nominal wind regimes, the temporal constraint should be less than 30 minutes and the spatial constraint within 25 km.
2. Set the calibration reference data source (e.g. $i = 1$), i.e. $a_1 = 1$ and $b_1 = 0$ (e.g. in situ).



3. Estimate the variance common to these smaller scales, $r^2 = \langle \delta_1 \delta_2 \rangle$ (representativeness error). See [Spatial Variance](#) to compute r^2 , ensuring that s_1 , s_2 , and s_3 represent the higher-, intermediate-, and coarser-resolution systems, respectively (e.g. buoys, satellite, and NWP model).
4. Compute the calibration coefficients iteratively until they satisfy a selected convergence threshold.
5. Calibrate the original measurements s_i^{uncal} with the new calibration coefficients:

$$s_i = \frac{s_i^{\text{uncal}} - b_i}{a_i}, \quad \forall i \in \{1, 2, 3\}.$$

This is an inverse calibration. At the first iteration the scaling coefficients are one and the bias coefficients are zero.

6. Apply the 4σ test to exclude possible outliers.
7. Estimate the first- and second-order moments and the covariances using the calibrated measurements for every combination of $i, j = 1, 2, 3$:

$$\begin{aligned} M_i &= \langle s_i \rangle, \\ M_{ij} &= \langle s_i s_j \rangle, \\ C_{ij} &= M_{ij} - M_i M_j. \end{aligned} \tag{6.7}$$

8. Estimate the true common variance T :

$$T = \frac{C_{12}C_{13}}{C_{23}} - r^2. \tag{6.8}$$

9. Estimate the calibration coefficients:

$$\begin{aligned} a_2 &= \frac{C_{23}}{C_{13}}, & b_2 &= M_2 - a_2 M_1. \\ a_3 &= \frac{C_{13}}{T}, & b_3 &= M_3 - a_3 M_1. \end{aligned}$$

10. Check convergence of $a_{2,3}$ and $b_{2,3}$:
 - $|a_i^k - a_i^{k-1}| \leq 10^{-6}$ for all $i \in \{2, 3\}$.
 - $|b_i^k - b_i^{k-1}| \leq 10^{-6}$ for all $i \in \{2, 3\}$, where k is the iteration index.
11. Iterate until convergence.
12. Estimate the measurement-error variances of the three sources:

$$\begin{aligned} \epsilon_1^2 &= \frac{C_{11}}{a_1^2} - T - r^2, \\ \epsilon_2^2 &= \frac{C_{22}}{a_2^2} - T - r^2, \\ \epsilon_3^2 &= \frac{C_{33}}{a_3^2} - T. \end{aligned} \tag{6.9}$$

Metrics

- $|a_i^k - a_i^{k-1}|$ for all $i \in \{2, 3\}$.
- $|b_i^k - b_i^{k-1}|$ for all $i \in \{2, 3\}$, where k is the iteration index.

Outputs

- Calibration coefficients: a_i, b_i for all $i \in \{1, 2, 3\}$.
- Measurement-error variances: ϵ_i^2 for all $i \in \{1, 2, 3\}$.

The error variances above are provided at the scale of the intermediate resolution system (e.g. satellite). They can also be computed at the scale of the coarser-resolution system (e.g. NWP model), using:

$$\begin{aligned}\epsilon_{\text{NWP},1}^2 &= \epsilon_{\text{SAT},1}^2 + r^2, \\ \epsilon_{\text{NWP},2}^2 &= \epsilon_{\text{SAT},2}^2 + r^2, \\ \epsilon_{\text{NWP},3}^2 &= \epsilon_{\text{SAT},3}^2.\end{aligned}\tag{6.10}$$

The labels SAT and NWP indicate the intermediate- and coarser-resolution systems, respectively.

6.2.3 One-to-One Validation

One-to-one (1-to-1) validation assesses the SAR-derived parameter against individual reference sources through linear correlation analysis. This approach is performed independently for each validation source, providing direct pairwise comparisons between SAR observations and reference data.

One-to-one validation is particularly useful when TC is not feasible due to insufficient collocated triplets. This may occur, for example, if the SAR Local Time Ascending Node (LTAN) does not align with other altimeter or scatterometer platforms and the region lacks in-situ sampling from buoys or other point-based platforms. In such cases, validation against NWP models or other available reference sources provides an essential quality assessment of the SAR retrievals.

Purpose

Assess the retrieval performance of SAR retrievals against a reference validation source.

Input

- SAR-derived parameter.
- NWP-model parameter or any other available validation source.

Procedure

- Collocate the two sources in space and time. A spatial constraint within 25 km and a temporal constraint within 30 minutes are recommended. For NWP models, collect hourly forecasts if available; otherwise interpolate model parameters bilinearly in space and time to the SAR wind-retrieval/vector cell and acquisition time.
- For circular variables such as wind direction, compute the differences $\Delta\phi$ and map the resulting angular differences to $[-180^\circ, 180^\circ]$:

$$\begin{aligned}\Delta\phi_{\text{raw}} &= \phi_{\text{SAR}} - \phi_{\text{VS}}, \\ \Delta\phi &= \Delta\phi_{\text{raw}} - 360^\circ \quad \text{if } \Delta\phi_{\text{raw}} > 180^\circ, \\ \Delta\phi &= \Delta\phi_{\text{raw}} + 360^\circ \quad \text{if } \Delta\phi_{\text{raw}} < -180^\circ.\end{aligned}\tag{6.11}$$

- Compute bias for all components.
- Compute root mean square difference (RMSD) for all components.
- For vector quantities, compute vector RMSD (vRMSD).

Metrics

- Non-vector quantities:
 - Bias: $b = \langle VAR_{\text{SAR}} - VAR_{\text{VS}} \rangle$.
 - $RMSD = \sqrt{\langle (VAR_{\text{SAR}} - VAR_{\text{VS}})^2 \rangle}$.
 - Pearson correlation coefficient:

$$r = \frac{\langle (VAR_{\text{SAR}} - \overline{VAR}_{\text{SAR}})(VAR_{\text{VS}} - \overline{VAR}_{\text{VS}}) \rangle}{\sigma_{\text{SAR}}\sigma_{\text{VS}}}.$$

- Vector quantities (e.g. wind vector):

- $b_w = \langle w_{\text{SAR}} - w_{\text{VS}} \rangle$.
- $RMSD_w = \sqrt{\langle (w_{\text{SAR}} - w_{\text{VS}})^2 \rangle}$.
- $vRMSD = \sqrt{\langle (u_{\text{SAR}} - u_{\text{VS}})^2 + (v_{\text{SAR}} - v_{\text{VS}})^2 \rangle}$.
- Pearson correlation coefficient:

$$r_w = \frac{\langle (w_{\text{SAR}} - \bar{w}_{\text{SAR}})(w_{\text{VS}} - \bar{w}_{\text{VS}}) \rangle}{\sigma_{\text{SAR}} \sigma_{\text{VS}}}.$$

- Circular variables (e.g. wind direction):

- $b_\phi = \langle \mathbf{mod}(\phi_{\text{SAR}} - \phi_{\text{VS}}, 360) \rangle$.
- $RMSD_\phi = \sqrt{\langle \mathbf{mod}(\phi_{\text{SAR}} - \phi_{\text{AS}}, 360)^2 \rangle}$.
- Jammalamadaka–Sarman circular-circular correlation coefficient:

$$\rho_c = \frac{\langle \sin(\phi_{\text{SAR}} - \bar{\phi}_{\text{SAR}}) \sin(\phi_{\text{VS}} - \bar{\phi}_{\text{VS}}) \rangle}{\sqrt{\langle \sin^2(\phi_{\text{SAR}} - \bar{\phi}_{\text{SAR}}) \rangle \langle \sin^2(\phi_{\text{VS}} - \bar{\phi}_{\text{VS}}) \rangle}}.$$

Here, angle brackets denote the expected-value operator, **mod** denotes modulus, and *VS* denotes validation source. *VAR* is the validation variable of interest. Formulas with subscript *w* apply to both components (*u*, *v*) and speed (*U*) for vector quantities.

Output

- Non-vector quantities:
 - Bias: b .
 - $RMSD$.
 - Pearson correlation coefficient: r .
- Vector quantities:
 - Bias for each component, speed, and direction: b_u, b_v, b_U, b_ϕ .
 - $RMSD$ for each component, speed, and direction: $RMSD_u, RMSD_v, RMSD_U, RMSD_\phi$.
 - $vRMSD$.
 - Pearson correlation coefficient for each component and speed: r_u, r_v, r_U .
 - Jammalamadaka–Sarman circular-circular correlation coefficient for direction: $\rho_{c,\phi}$.

6.2.4 Spectral Analysis

Purpose

Spectral analysis is a validation tool used to assess the spatial scales resolved by the retrievals (*Vogelzang et al. (2011)*).

Inputs

- M sets of N wind-component samples each, denoted $w_{ij}, i = 1, \dots, M, j = 1, \dots, N$.

Procedure

- Check for missing values in each set. Remove sets with missing values or fill the gaps using an interpolation procedure.
- Compute the power spectral density (PSD):

$$\psi_{w,j}^i = \psi_{w,k_j}^i = \begin{cases} \frac{\Delta}{N} |\hat{w}(k_j)|^2, & j = 0 \text{ or } j = \frac{N}{2}, \\ \frac{\Delta}{N} \left(|\hat{w}(k_j)|^2 + |\hat{w}(k_{-j})|^2 \right), & j = 1, \dots, \frac{N}{2} - 1, \end{cases} \quad i = 1, \dots, M. \quad (6.12)$$



Here, $\hat{w}$ is the Fast-Fourier transform (FFT) of the (vector) component w , Δ is the spacing, and k_j is the wavenumber.

- Average the PSD from each set:

$$\Psi_{w,j} = \frac{1}{M} \sum_{i=1}^M \psi_{w,j}^i. \quad (6.13)$$

PSD units are the squared units of the analyzed variable multiplied by the spatial sampling interval Δ (in metres). For wind components and LoS surface currents, whose units are m s^{-1} , PSD units are $\text{m}^3 \text{s}^{-2}$; for SWH, whose units are m, PSD units are m^3 .

Metrics

In accordance with the Guide to the Expression of Uncertainty in Measurement (GUM) Type A evaluation principles for spectral estimators, the standard uncertainty $u(\Psi_{w,j})$ of the averaged PSD is driven by the number of independent scenes M . Assuming a χ^2 distribution with $2M$ degrees of freedom for each spectral bin j , the relative standard uncertainty is:

$$\frac{u(\Psi_{w,j})}{\Psi_{w,j}} = \frac{1}{\sqrt{M}}. \quad (6.14)$$

Outputs

- PSD for each wind component: $\Psi_{u,j}, \Psi_{v,j}$.
- Associated standard-uncertainty profiles derived from (6.14).

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¹⁰ <https://doi.org/10.1029/2010JC006640>

CHAPTER SEVEN

UNCERTAINTY ANALYSIS

7.1 Introduction

The “Evaluation of measurement data – GUM” (*BIPM et al., n.d.-a*), or the “GUM bible”, is a foundational international document that provides a conceptual and mathematical framework for identifying sources of uncertainty, quantifying them in a consistent way (as standard deviations), combining them into a single uncertainty figure, and reporting the result and its uncertainty clearly and comparably across disciplines and countries. Focusing on evaluation and expression of uncertainty in measurement it is an important component of the science of measurement and application, known as metrology (e.g., *BIPM et al., n.d.-b*).

The GEO is an international partnership of governments and organizations acting as a policy and coordination body to promote global sharing and use of EO data, for example to support work on climate, disaster management, agriculture, water, and ecosystems. The GEOSS is the global, distributed infrastructure that GEO is building and coordinating. Rather than being a single system, it connects many independent EO systems – such as satellites, ground networks, and ocean buoys – from different countries and agencies, and aims to make them interoperable, accessible through common standards and interfaces, and accompanied by information about data quality and uncertainty. As such, GEO is the partnership, and GEOSS is the “system of systems” that GEO oversees.

To support its mission, GEOSS needs a metrological framework. As such, the 2010 endorsement of the QA4EO by the CEOS (part of GEOSS), has defined guidelines for EO applications based on principles adapted from the metrology community (*Woolliams et al. 2025a*). QA4EO aims to ensure that EO data are accessible, of known and documented quality, traceable to reference standards, in particular the SI, and suitable for user application (“fit for purpose”).

This section applies the GUM principles to SAR data and reviews the existing frameworks supporting metrology: the VIM (*BIPM et al., n.d.-b*), the GUM (*BIPM et al., n.d.-a*), and QA4EO.

7.2 The GUM Principles: A review

7.2.1 The International Vocabulary of Metrology (VIM)

The International Vocabulary of Metrology (VIM; *BIPM et al., n.d.-b*) defines and organizes key concepts in metrology to ensure global consistency in measurement science. It provides standardized terminology covering quantities, units, and measuring instruments, with a focus on measurement uncertainty rather than error.

The document explains critical properties of measuring devices, such as precision, stability, and sensitivity, and highlights the importance of traceability and standards for reliability.

Its applications span various fields, including physics, chemistry, biology, medicine, and emerging areas like forensic science and food technology. Harmonization of terms is emphasized to ensure coherence in the various scientific, engineering, regulatory, and quality assurance fields.

7.2.2 Guide to the Expression of Uncertainty in Measurement (GUM)

The GUM establishes general rules for evaluating and expressing uncertainty in measurement that are intended to be applicable to a broad spectrum of measurements (*BIPM et al., n.d.-a*). The guide was developed to solve the challenge of different laboratories and disciplines using different methods and terminology for uncertainty, which made measurement results difficult to compare.



The central idea of the GUM is to treat all components of uncertainty in the same mathematical way, regardless of whether they arise from random variation or systematic effects. Every component is represented by a probability distribution and a standard uncertainty. To obtain standard uncertainties, the GUM distinguishes between two kinds of evaluation. “Type A” evaluations use statistical analysis of repeated measurements, whereas “Type B” evaluations use all other available information, such as instrument specifications, calibration certificates, or expert judgment. The results of both evaluation types can be converted into standard uncertainties so they can be used on the same terms.

Measurements are described by a mathematical model that links the quantity being measured (the measurand) to various input, or influence, quantities. Once standard uncertainties of the influence quantities are estimated, they are combined – taking into account any correlations – using standard rules for propagating variances. This yields the combined standard uncertainty of the measurand, taken as the estimated standard deviation associated with the result. An expanded uncertainty is obtained by multiplying the standard uncertainty by a coverage factor k . The purpose of the expanded uncertainty is to provide an interval about the result of a measurement that may be expected to encompass a large fraction of the distribution of values that could reasonably be attributed to the measurand. The value of k and how it was chosen must always be stated.

A key emphasis of the GUM is transparency. An uncertainty statement should clearly describe which inputs were considered, how each uncertainty component was obtained, what assumptions were made, and how the components were combined. The document supplements this framework with definitions, statistical background, guidance on practical evaluation, and worked examples. In doing so, it provides a common language and method that allow measurement results and their uncertainties to be understood and compared across borders and domains.

7.2.3 Quality Assurance Framework for Earth Observation (QA4EO)

GEOSS depends on interoperable and quality-assured data to derive comprehensive information about climate and environment, as well as, e.g., food, water and energy security. To achieve interoperability across systems and datasets, common and well-defined data and metadata formats are needed, but also metrological principles such as traceability to SI, documented uncertainty analysis and comparison between different measurement references must be followed, and are key to the QA4EO reports (*Woolliams et al. 2025a*).

Beside *BIPM et al., n.d.-b* and *BIPM et al., n.d.-a*, QA4EO also builds on EO-metrology collaboration in the EU FP7 and Horizon 2020, ESA and Copernicus projects and, in particular, the FIDUCEO project (*Mittaz et al. 2019*).

As summarized in (*Woolliams et al. 2025b*), three basic metrology principles generally apply to EO data:

1. **Traceability:** Metrological traceability is a property of a measurement that relates the measured value to a stated metrological reference through an unbroken chain of calibrations or comparisons. It requires, for each step in the traceability chain, that uncertainties are evaluated and that methodologies are documented.
2. **Uncertainty Analysis:** Uncertainty analysis is the review of all sources of uncertainty and the propagation of that uncertainty through the traceability chain. It is based on the suite of documents prepared by the JCGM and together known as the GUM.
3. **Comparison:** Metrologists validate uncertainty analysis and confirm traceability through comparisons. The Mutual Recognition Arrangement (a formal arrangement to confirm the degree-of-equivalence between SI realisations and disseminations in different countries) requires regular, formal international comparisons that are conducted under strict rules.

The uncertainty analysis, as outlined in the GUM, should follow these principles.

ESA has initially applied the terms FRM, FDR and TDP to describe metrologically rigorous observations of specific relevance to space-based observations (*Woolliams et al. 2025b*). The terms are increasingly being used by the broader EO community, although they have not yet been formally endorsed. The terms are defined in *Woolliams et al. 2025b*, as follows:

- **Fundamental Data Record (FDR):** An FDR is a record, of sufficient duration for its application, of uncertainty-quantified sensor observations calibrated to physical units and located in time and space, together with all ancillary and lower-level instrument data used to calibrate and locate the observations and to estimate uncertainty.
- **Thematic Data Product (TDP):** A TDP is a record, of sufficient duration for its application, of uncertainty-quantified retrieved values of a geophysical variable, along with all ancillary data used in retrieval and uncertainty estimation.

- **Fiducial Reference Measurement (FRM):** An FRM is a suite of independent, fully characterised, and traceable (to a community agreed reference, ideally SI) measurements of a satellite relevant measurand, tailored specifically to address the calibration/validation needs of a class of satellite borne sensors and that follow the guidelines outlined by the GEO/CEOS QA4EO.

As highlighted by *Woolliams et al. 2025b*, “TDPs provide higher level products that have been processed from FDRs, through algorithms which also often combine information from other FDRs (e.g., from other satellite sensors) or from external information (such as reanalysis models and/or certain non-satellite data), along with such additional information”. Guidelines for assessing observational systems with respect to FRMs have been established by CEOS.

In order to establish a systematic process for generating uncertainty budgets, and ensuring that EO data are interoperable, temporally stable, and fit for scientific, operational and decision-making purposes, the QA4EO and EO projects acknowledged in *Woolliams et al. 2025c* have adapted the stages, suggested in GUM, required to support the development of a measurement model and from that an uncertainty analysis. These EO specific stages of uncertainty analysis are given by *Woolliams et al. 2025c* as follows:

- Step 1: Define the measurand and measurement model.
- Step 2: Establish the traceability with a diagram.
- Step 3: Evaluate each source of uncertainty and fill out an effects table.
- Step 4: Calculate the data product and uncertainties.
- Step 5: Record information about the uncertainty analysis for long term data preservation purposes and summarise for today’s users.

These steps are discussed in detail in *Woolliams et al. 2025c*.

7.3 Product-specific uncertainty estimation

7.3.1 Azimuth wavelength cutoff

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.2 MACS

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.3 iMACS

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.4 DCA

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.5 Texture-based azimuth cutoff

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.6 RVL

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.7 Inverse Wave Age

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.8 Wind products

7.3.8.1 Wind speed, wind stress, and hub height wind speed

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.9 Sea-state products

Total significant wave height; mean period wind-sea; first and second moment wave period; wave height for the swell-dominant and swell-secondary systems; significant wave height wind-sea; wave direction; and wave directional energy spectrum.

- Description of measurand and measurement model
- Traceability matrix



- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.10 Line-of-sight current products

7.3.10.1 Line-of-sight wave Doppler and line-of-sight surface current component

- Description of measurand and measurement model
- Traceability matrix
- Evaluation of effects
- Calculation of associated uncertainties
- Documentation and outputs

7.3.11 Summary

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